# Phase 1 – Core MVP (Universal Foundation)

**Universal (Schools + Companies)**

- User login / register
- Clock In & Out (GPS + QR)
- Offline clock-in/out with auto-sync
- Attendance history (personal log)
- Basic notifications (success, failure, out-of-zone alerts)
- Admin: staff/student management (add, edit, delete)
- Admin: branch/location creation + QR code generation
- Admin: attendance logs (daily/weekly)
- Admin: real-time status (on time, late, absent)
- Admin: basic shift management (morning/evening)
- Admin: exportable reports (Excel/PDF)

# Phase 2 – Flexibility & Control

**Universal (Schools + Companies)**

- Proxy clock-in (with GPS + photo + consent proof)
- Emergency clock-in (with reason, pardons)
- Early leave option (with reason)
- Multi-institution account switching (for teachers & part-time staff)
- Penalty system (lateness, early leave, absence, linked to payroll logic)
- Department/Category management (companies: HR, IT | schools: Classes, Subjects)
- Approve/reject emergency clock-ins
- Advanced reporting (filter by branch, shift, department/class)

# Phase 3 – Advanced & Differentiators

**For Companies**

- Payroll system integration (attendance → salary adjustments)
- Employee behavior analytics (latecomer trends, absentee history)
- Audit trail (who edited attendance/penalties)
- Custom branding / white-label option

**For Schools**

- Student attendance profiles (with photo + QR)
- Class creation & management
- ID card generator (with QR code + school branding)

- Parent notifications (SMS/Email for absences/lateness)

**Shared (Both)**

- Mobile push notifications (shift reminders, late alerts)
- Shift query feature (“When is my next shift/class?”)
- Multi-language support
- API integrations (for HR, ERP, or school management systems)

## Phase 1 (Core MVP) – 1 Month Plan

# Week 1 – Foundations

**Mobile Dev**

- Set up React Native/Flutter project.
- Implement user registration & login (JWT auth).
- Build basic UI (login, dashboard, attendance screen).

**Web/Backend Dev**

- Set up backend (Node.js + Express + MongoDB/Postgres).
- Implement authentication APIs (register, login).
- Create database schemas (users, attendance, institutions, branches).
- Set up admin dashboard boilerplate (React/Next.js).

# Week 2 – Attendance Core

**Mobile Dev**

- Implement GPS-based clock-in/out.
- Integrate QR scanner for location verification.
- Store clock-in/out logs locally (SQLite/AsyncStorage).

**Web/Backend Dev**

- Build API endpoints for clock-in/out.
- Implement attendance logs storage in DB.
- Admin dashboard: view attendance logs (daily/weekly).

- Generate QR codes for branches.

## Week 3 – Offline & History

**Mobile Dev**

- Implement offline clock-in/out (local storage → auto-sync when online).
- Build personal attendance history (view past logs).
- Add basic notifications (success/failure, out-of-zone).

**Web/Backend Dev**

- Handle offline sync logic in backend.
- Admin dashboard: real-time status (on time, late, absent).
- Implement staff/student management (add, edit, delete).

## Week 4 – Shifts & Reports

**Mobile Dev**

- Basic shift handling (morning/evening selection).
- Polish UI/UX (error messages, loading states, validations).
- Final testing & bug fixing.

**Web/Backend Dev**

- Basic shift management (create, assign).
- Export attendance reports (Excel/PDF).
- Admin dashboard refinements (filters, search).
- Deployment (host backend & dashboard, publish mobile beta).

## Deliverables at End of Month

- **Mobile App:**
    - Login/Register
    - GPS + QR clock-in/out
    - Offline clock-in/out with auto-sync
    - Personal attendance history
    - Basic notifications

- **Web Dashboard:**
  - Staff/student management
  - Branch/location setup + QR codes
  - Attendance logs (daily/weekly)
  - Real-time status (on time, late, absent)
  - Basic shifts (morning/evening)
  - Export reports (Excel/PDF)

# Week 1 – Foundations

**Goal:** Core setup (projects, auth, data models, base dashboard).

- [ ] **Mobile (Dev A)**
  - Set up project (React Native/Flutter).
  - Implement login & register (connect to backend auth).
  - Create basic screens: Login, Dashboard, Clock In/Out placeholder.
- [ ] **Web/Backend (Dev B)**
  - Set up backend (Node.js + Express + MongoDB).
  - Implement user registration & login APIs (JWT).
  - Create DB schemas (Users, Institutions, Branches, Attendance).
  - Set up Admin Dashboard skeleton (React/Next.js).

**Demo at end of week:**

- User can register/login (mobile + web).
- Admin can log into dashboard (even if empty).

# Week 2 – Attendance Core

**Goal:** Make clock-in/out functional with GPS + QR.

- [ ] **Mobile (Dev A)**
  - Implement GPS clock-in/out.
  - Integrate QR scanner for location verification.
  - Send attendance logs to backend API.
- [ ] **Web/Backend (Dev B)**
  - Build attendance API endpoints (clock-in/out).
  - Store logs in DB (with user, timestamp, GPS, QR ID).
  - Admin dashboard: display attendance logs (daily/weekly).

    - Generate QR codes for each branch/location.

**Demo at end of week:**

- Staff can clock-in/out via mobile → appears in DB.
- Admin can view logs in dashboard.
- QR codes working.

# Week 3 – Offline & Logs

**Goal:** Ensure reliability (offline mode, logs visible).

- ☐ **Mobile (Dev A)**
    - Implement offline clock-in/out (local storage).
    - Auto-sync to server when online.
    - Attendance history screen (user's past logs).
- ☐ **Web/Backend (Dev B)**
    - Handle sync in backend (accept offline logs with timestamps).
    - Admin dashboard: real-time status (on time, late, absent).
    - Staff/student management (CRUD).

**Demo at end of week:**

- Staff clocks in offline → syncs when internet restored.
- Admin sees synced logs in dashboard.
- Admin can add/remove staff → reflected in app.

# Week 4 – Shifts & Reports

**Goal:** Finish Phase 1 with shifts + reporting.

- ☐ **Mobile (Dev A)**
    - Add basic shift handling (morning/evening).
    - Finalize UI polish (buttons, error states, confirmations).
    - Test full flow end-to-end.
- ☐ **Web/Backend (Dev B)**
    - Implement shift management (create, assign).
    - Export attendance reports (Excel/PDF).
    - Final testing + deployment (backend + dashboard live, mobile beta).

**Demo at end of week:**

- Staff can clock-in/out based on shifts.
- Admin can assign shifts + export reports.
- Full system ready for pilot testing.